

Meta-analysis

Master of statistics

KU Leuven

Effect size calculation

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Steps in a meta-analysis

1. Formulating research questions
2. Carefully plan your study
3. Systematic search for studies
4. Selection of relevant studies
5. Extracting relevant data (outcomes, study characteristics, study quality)
6. Converting data to a common effect size measure
7. Combining and comparing the effect sizes
8. Looking for moderator variables
9. Discussing and reporting



The use of effect sizes in meta-analysis

- The effect size summarizes the results of the primary studies
- It is the dependent variable in the meta-analysis
- Often a standardized effect size measure is used, allowing comparison over studies
- There are several possible effect size indices: e.g.
 - the standardized mean difference
 - The correlation coefficient
 - The odds ratio
- Before being combined in a meta-analysis, findings from primary studies are converted to a common measure



Effect size calculation

a. why using effect sizes?



The interpretation of p -values

Example: testing the difference in the size of tumors in an experimental and a control group

What would you conclude if

$$\rightarrow p = .11?$$

$$\rightarrow p < .0001?$$



The interpretation of p -values

Misconceptions:

- failure to reject the null hypothesis implies no effect
- a statistically significant p -value implies a large effect



$$\text{Test of Significance} = \text{Size of Effect} \times \text{Size of Study}$$

Rosenthal, R. (1991). *Meta-analytic procedures for social research*. Newbury Park, CA: Sage



Test of Significance = Size of Effect × Size of Study

Example: two sample t -test (assuming common variance)

$$t = \frac{\bar{X}_E - \bar{X}_C}{s_p \sqrt{\frac{1}{n_E} + \frac{1}{n_C}}}$$

$$g = \frac{\bar{X}_E - \bar{X}_C}{s_p}$$

$$\Rightarrow t = g * \left(\sqrt{\frac{1}{n_E} + \frac{1}{n_C}} \right)^{-1}$$



Stress on effect sizes

E.g.,

- ▮ APA Publication Manual
- ▮ Guidelines of journals

The effect size

- ▮ Expresses the magnitude of the ‘effect’ or relationship,
- ▮ independently of the study size



'Effect' sizes?

- Expressing effects in (quasi-)experimental studies
 - Comparing experimental & control groups
 - Comparing one group under several conditions
- Expressing association in non-experimental studies
 - Comparing existing groups (e.g., male vs. female)
 - Expressing association between continuous variables (e.g., relation between school size and performance)
 - Expressing reliability of a test (e.g., Cronbach's alpha; combined in 'Validity generalization' study)
- Describing one single variable (e.g., prevalence rates, means, ...)



Association $\neq$ Causation !

Check the internal validity of the design!

(are there confounding variables?)

Pay attention with the interpretation of your results!



Interpretation of effect sizes

Unstandardized (e.g., a mean difference)

- if the units of measurement are practically meaningful (e.g., number of cigarettes smoked per day; length in centimetres)
- if the scores are comparable over the primary studies



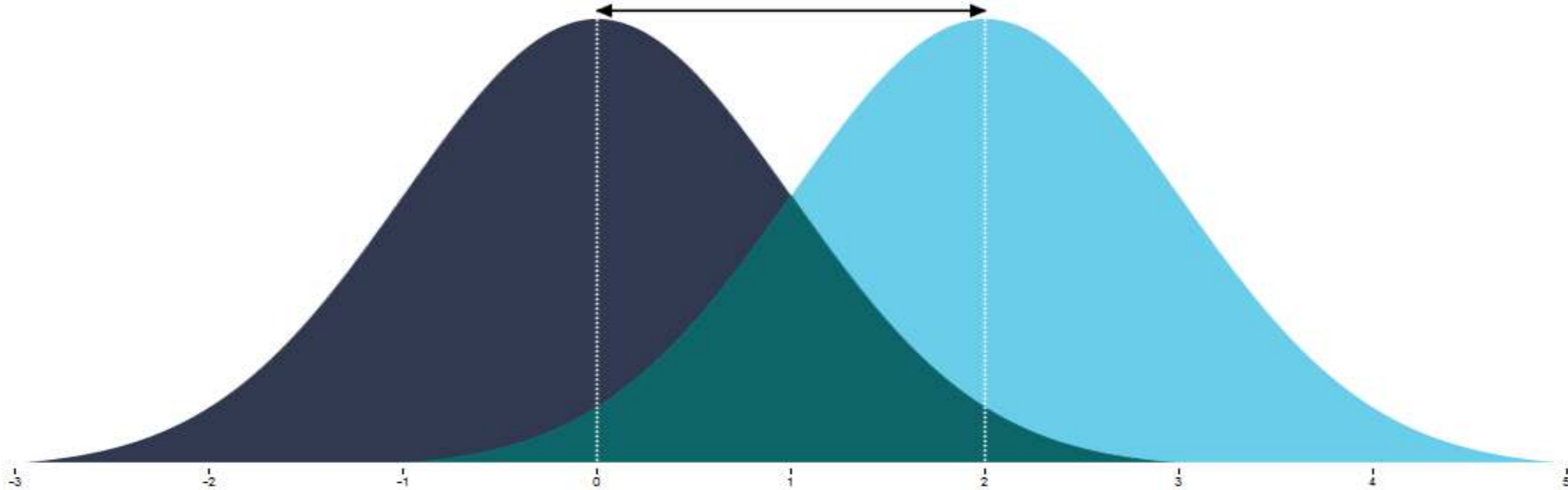
Interpretation of standardized effect sizes

<http://rpsychologist.com/d3/cohend/>

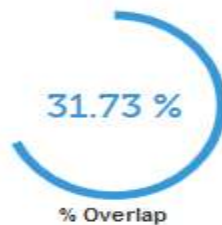
<http://rpsychologist.com/d3/correlation/>

Slide me ⚙

Cohen's d: 2



Interpretation



Interpretation of standardized effect sizes

(Cfr. Cohen, 1969; 1988; "there is a certain risk inherent in offering conventional operational definitions for those terms for use in power analysis in as diverse a field of inquiry as behavioral science" (p. 25).)

- small: $d = 0.20$ ($r = .10$; $r^2 = .01$)
the size of the difference between
15- and 16-year-old girls
- medium: $d = 0.50$ ($r = .243$; $r^2 = .059$)
visible to the naked eye
14- and 18-year-old girls
- large: $d = 0.80$ ($r = .371$; $r^2 = .138$)
grossly perceptible
13- and 18-year-old girls



When small effects are impressive

Prentice, D. A., & Miller, D. T. (1992). When small effects are impressive. *Psychological Bulletin*, 112, 160-164.

Evaluating the “impressiveness” of the effect by:

- Minimal manipulations of independent variable (e.g., cookie experiment)
- Choosing a supposedly stable dependent variable (e.g., effect of physical attractiveness of defendants on judgments of guilt and severity of punishment)



Size of the effect $\neq$ practical significance

Practical significance depends on

- The costs of the treatment
- The severity of the problem under investigation
- The extent of the problem
- ...



Effect size calculation

a. why using effect sizes?

b. a metric for each situation



Independent variable	Dependent variable		
	Dichotomous	Nominal	Ordinal/interval
	RD RR Φ OR		g_{IG} Glass's Δ g_{gain} $g_{gain IG}$ nonparametric r_{pb}
	Nominal	Contingency Measures Goodman-Kruskal's Tau Uncertainty coefficient Cohen's Kappa	Multiple g's η^2 ICC
Ordinal/interval			Pearson's r Spearman's ρ Kendall's τ / Somer's D Gamma coefficient Weighted Kappa



Dichotomous independent-dichotomous dependent variable

Predictive test	Final exam		
	1	0	
1	130	20	150
0	30	20	50
	160	40	200



Dichotomous independent-dichotomous dependent variable

Predictive test	Final exam		
	1	0	
1	130 (87 %)	20 (13 %)	150 (100 %)
0	30 (60 %)	20 (40%)	50 (100 %)
	160	40	200

1. Risk difference: $.87 - .60 = .27$
2. Relative risk: $.87 / .60 = 1.45$
3. Phi: $(130 \times 20 - 20 \times 30) / \sqrt{(150 \times 50 \times 160 \times 40)} = 0.29$
4. Odds ratio: $(130 \times 20 / 20 \times 30) = 4.33$
5. Log odds ratio = $\log(4.33) = 1.47$

Independent variable	Dependent variable		
	Dichotomous	Nominal	Ordinal/interval
	Dichotomous	RD RR Φ OR	g_{IG} Glass's Δ g_{gain} $g_{gain\ IG}$ nonparametric r_{pb}
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Dichotomous independent-continuous dependent variable

1. Independent groups, homogeneous variance:

$$g = \frac{\bar{X}_E - \bar{X}_C}{s_p}$$

2. Independent groups, heterogeneous variance:

$$\text{Glass's } \Delta = \frac{\bar{X}_E - \bar{X}_C}{s_C}$$

3. Repeated measures^{s_C}(one group):

4. Repeated measures^{s_D} or $g_{\text{gain}} = \frac{D}{s_D}$ or $g = \frac{\bar{X}_{\text{post}} - \bar{X}_{\text{pre}}}{s_{\text{pre}}}$ (independent groups):

5. $g_{\text{gain-IG}} = \frac{\bar{X}_{\text{postE}} - \bar{X}_{\text{preE}}}{s_{\text{preE}}} - \frac{\bar{X}_{\text{postC}} - \bar{X}_{\text{preC}}}{s_{\text{preC}}}$ Nonparametric & robust measures (e.g., difference between medians)

6. r_{pb}



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	Illness		
	Better	Same	Worse
Experimental	10	5	2
Control	4	7	3

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	RR		Glass's Δ
	Φ		g_{gain}
	OR		$g_{gain IG}$
			nonparametric
			r_{pb}
Nominal		Contingency Measures	Multiple g's
		Goodman-Kruskal's	η^2
		Tau	ICC
		Uncertainty coefficient	
		Cohen's Kappa	
Ordinal/interval			Pearson's r
			Spearman's ρ
			Kendall's τ / Somer's D
			Gamma coefficient
			Weighted Kappa



Independent variable	Dependent variable		
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Regression models

1. Y continuous: $Y_i = a + bX + e_i$

1. X continuous: b estimated by $\frac{s_{XY}}{s_X}$

2. X dichotomous (1 = experimental, 0 = control),
b estimated by $\bar{y}_E - \bar{y}_C$

2. Y dichotomous: $\text{Logit}(P(Y=1)) = \text{Log odds} = a + bX$,

If X dichotomous, b is a log odds ratio (LOR)

⇒ r, g, LOR are (standardized) regression coefficients!



Effect size calculation

- a. why using effect sizes?**
- b. a metric for each situation**
- c. sampling distribution of effect sizes**

**Why is this
important?**



The standardized mean difference

$$g \sim N(\delta, \sigma_g^2)$$

$$\sigma_g^2 \cong \frac{n_E + n_C}{n_E n_C} + \frac{\delta^2}{2(n_E + n_C)}$$

$$\hat{\sigma}_g^2 \cong \frac{n_E + n_C}{n_E n_C} + \frac{g^2}{2(n_E + n_C)}$$

If group sizes equal and relatively large:

$$\hat{\sigma}_g^2 \cong \frac{4}{N}$$



The correlation coefficient

$$r \sim N(\rho, \frac{(1-\rho^2)^2}{N})$$

But:

- only approximately normal
- Bias: $E(r) \cong \rho - \rho(1-\rho^2)/(2N-2)$
- Sampling variance strongly dependent on ρ

⇒ Fisher's Z_r transformation

$$Z_r = 0.5 \log \left(\frac{1+r}{1-r} \right)$$

$$\sigma_{Z_r}^2 = \frac{1}{N-3}$$

Afterwards backward transformation: $r = \frac{e^{2Z_r} - 1}{e^{2Z_r} + 1}$

The odds ratio (ad/bc)

Predictive test	Final exam	
	1	0
1	a	b
0	c	d

- Asymmetric distribution
- Complex standard error formula

⇒ Log odds ratio

- Normal sampling distribution

- $$\sigma_{LOR}^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$$



Choosing a measure

1. Design and measurement level
2. Comparability over studies (e.g., by standardization)
3. Assumptions
4. Popularity
5. Simplicity of measure
6. Directional effect size
7. Simplicity of sampling distribution

Effect size calculation

a. why using effect sizes?

b. a measure for each situation

**c. sampling distribution of
effect sizes**

**d. comparability of effect size
measures**



Comparability of effect sizes

Example: g_{IG} vs. g_{gain} :

$$\delta_{IG} = \frac{\mu_E - \mu_C}{\sigma}$$

$$\delta_{gain} = \frac{\mu_D}{\sigma_D} = \frac{\mu_{post-pre}}{\sigma_{post-pre}}$$

$$= \frac{\mu_{post} - \mu_{pre}}{\sqrt{\sigma_{post}^2 + \sigma_{pre}^2 - 2\rho\sigma_{pre}\sigma_{post}}}$$

$$= \frac{\mu_{post} - \mu_{pre}}{\sigma\sqrt{2(1-\rho)}}$$

$$\delta_{gain} = \frac{\delta_{IG}}{\sqrt{2(1-\rho)}}$$



Comparability of effect sizes

1. Estimating different population parameters, e.g.,

$$g_{gain} = \frac{g_{IG}}{\sqrt{2(1-r)}}$$

2. Estimating with different precision, e.g.,
g vs. Glass's Δ

Effect size calculation

- a. why using effect sizes?**
- b. a measure for each situation**
- c. sampling distribution of effect sizes**
- d. comparability of effect sizes**
- e. calculation of effect sizes**

(based on slides retrieved on February 18
2015, from
<http://mason.gmu.edu/~dwilsonb/ma.html>)

The different formulas represent degrees of approximation to the ES value that would be obtained based on the means and standard deviations

Great

- direct calculation based on means and standard deviations
- algebraically equivalent formulas (t-test)
- exact probability value for a t-test
- approximations based on continuous data (correlation coefficient)

Good

- Results of one way ANOVA with 3 or more groups
- Results of ANCOVA
- Results of multiple regression analysis

Poor

- approximations based on dichotomous data



Methods of Calculating the Standardized Mean Difference

Direct Calculation Method

$$ES = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}}} = \frac{\bar{X}_1 - \bar{X}_2}{S_{pooled}}$$

Methods of Calculating the Standardized Mean Difference

Algebraically Equivalent Formulas:

$$ES = t \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$$

independent t-test

$$ES = \sqrt{\frac{F(n_1 + n_2)}{n_1 n_2}}$$

two-group one-way ANOVA

exact *p-values* from a *t*-test or *F*-ratio can be converted into *t*-value and the above formula applied

Methods of Calculating the Standardized Mean Difference

Close Approximation Based on Continuous Data -- Point-Biserial Correlation. For example, the correlation between treatment/no treatment and outcome measured on a continuous scale.

$$ES = \frac{2r}{\sqrt{1 - r^2}}$$

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Methods of Calculating the Standardized Mean Difference

Estimates of the Denominator of ES --
Pooled Standard Deviation

$$s_{pooled} = \sqrt{MS_{error}}$$
$$= \sqrt{\frac{MS_{between}}{F}}$$

one-way ANOVA >2 groups

Methods of Calculating the Standardized Mean Difference

Estimates of the Denominator of ES --
Pooled Standard Deviation

$$s_{pooled} = \sqrt{\frac{SS_B + SS_{AB} + SS_W}{df_B + df_{AB} + df_W}}$$

A two-way factorial ANOVA where B is the irrelevant factor and AB is the interaction between the irrelevant factor and group membership (factor A).



Methods of Calculating the Standardized Mean Difference

Estimates of the Denominator of ES --
Pooled Standard Deviation

$$s_{pooled} = \sqrt{\frac{MS_{error}}{1 - r^2} \cdot \frac{df_{error} - 1}{df_{error} - 2}}$$

ANCOVA, where r is the correlation between the covariate and the DV

The different formulas represent degrees of approximation to the ES value that would be obtained based on the means and standard deviations

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Methods of Calculating the Standardized Mean Difference

Approximations Based on Dichotomous Data

$$ES = \text{probit}(p_{\text{group}_1}) - \text{probit}(p_{\text{group}_2})$$

the difference between the probits
transformation
of the proportion successful in each group

converts proportion into a z-value



Methods of Calculating the Standardized Mean Difference

Approximations Based on Dichotomous Data

$$ES = \frac{\sqrt{3}}{\pi} (\log[\text{Odds - ratio}])$$

this represents the rescaling of the logged odds-ratio (see Sanchez-Meca et al 2004 Psychological Methods article)



Methods of Calculating the Standardized Mean Difference

Approximations Based on Dichotomous Data

$$ES = 2\sqrt{\frac{\chi^2}{N - \chi^2}}$$

chi-square must be based on a 2 by 2 contingency table (i.e., have only 1 df)

$$ES = \frac{2r}{\sqrt{1 - r^2}}$$

phi coefficient

Doing the calculations

- Online tool (Wilson): <http://www.campbellcollaboration.org/escalc/html/EffectSizeCalculator-Home.php>
- R, for instance the metafor-package or the esc-package (based on the online tool of Wilson)
- CMA
- MS Excel

Effect size calculation

- a. why using effect sizes?**
- b. a measure for each situation**
- c. sampling distribution of effect sizes**
- d. comparability of effect size measures**
- e. calculation of effect sizes**
- f. threats of effect size measures**

Threats of effect sizes

1. 'Bad data'
2. Measurement error
3. Artificial dichotomization
4. Imperfect construct validity
5. Biased estimator $g' = g \left[1 - \frac{3}{4N - 9} \right]$
6. Range restriction

(Hunter, J. E., & Schmidt, F. L. (1990). *Methods of meta-analysis: Correcting error and bias in research findings*. Newbury Park, CA: Sage.)

